Reaction kinetics: basic concepts

Kinetics is the study of reaction rates and how they are affected by factors such as reactant concentrations, pressure, temperature, pH or enzyme activity.

Reaction rate

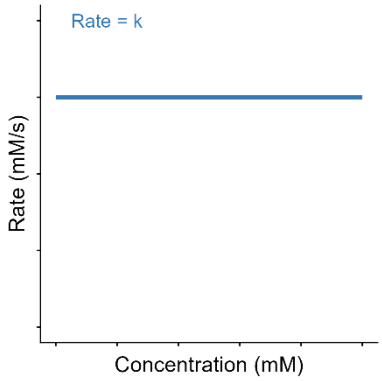
The **rate of a chemical reaction** is defined as the concentration of a product that is formed, or the concentration of a reactant that is consumed, in a unit of time.

Units in the International System are: $mol \cdot L^{-1} \cdot s^{-1}$.

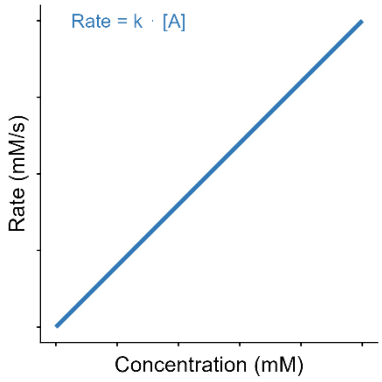
Order of reaction

The **order of a chemical reaction** refers to the relationship between the rate and the concentration of reactants.

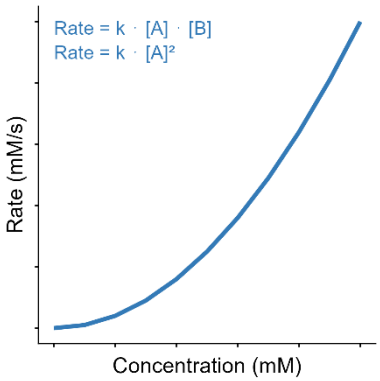
In a **zero-order** reaction, the rate is independent of the concentration of reactants.



In a **first-order** reaction, the reaction rate is directly proportional to the concentration of one reactant.



In a **second-order** reaction, the rate is dependent on the concentration of two different reactants (in a bisubstrate reaction) or on the square of the concentration of a single reactant (for example, in a dimerization reaction).



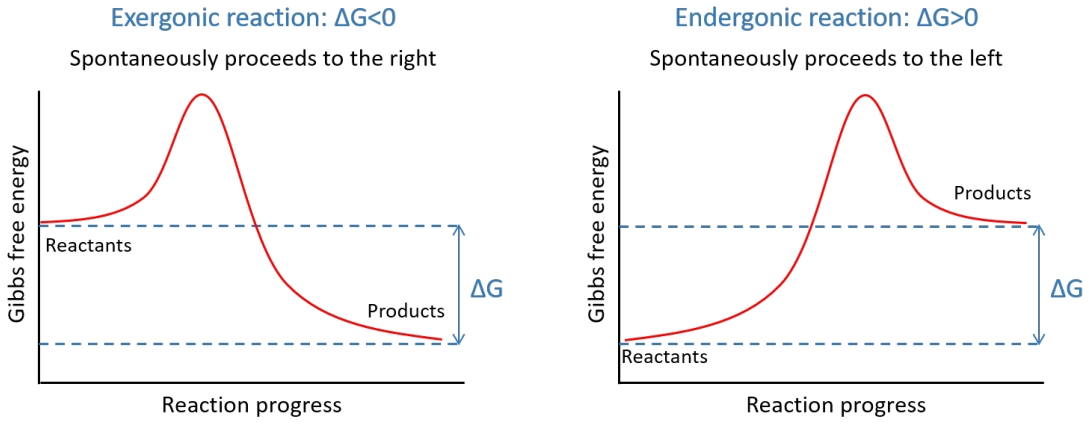
Why is this relevant to enzymes?

Enzyme-catalyzed reactions can show different kinetic behaviors depending on substrate concentration. At low substrate concentrations, the reaction rate is approximately proportional to [S], whereas at high substrate concentrations, the rate becomes independent of [S] because the enzyme is saturated.

Free energy change of a reaction

Gibbs free energy change (ΔG) determines the thermodynamically favorable direction of a reaction.

Reactions that have a ΔG < 0 (i.e. the products of the reaction have less free energy than the reactants) are **exergonic** (release energy) and, therefore, **spontaneous**.



Exergonic reactions (ΔG < 0) are thermodynamically favorable, whereas endergonic reactions (ΔG > 0) are not thermodynamically favorable in the forward direction.

Equilibrium

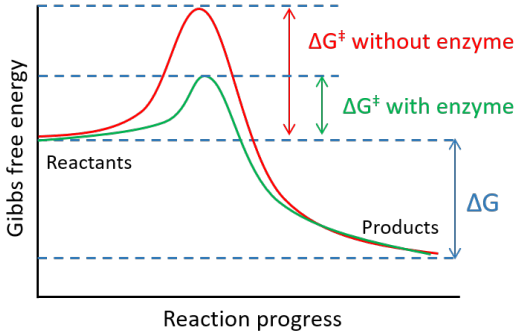
Chemical equilibrium is reached when the rates of the forward and reverse reactions are equal; therefore, reactants and products are in a state of dynamic equilibrium and their concentrations no longer change. At equilibrium, the concentrations of reactants and products remain constant, but they are not necessarily equal. **At equilibrium, ΔG = 0**.

Energy of activation and catalysis

Even exergonic reactions require an energy input to begin (**activation energy**). This energy is needed to reach the **transition state**.

The higher the activation energy, the slower the chemical reaction will be. Therefore, **a spontaneous reaction does not imply that the reaction will occur quickly**.

Most cellular reactions proceed at appreciable rates because **enzymes** lower their activation energies.



Activation energy is the energy required for a reaction to proceed. Enzymes are biological catalysts that lower the activation energy of a reaction.

Important: Enzymes speed up chemical reactions by lowering the activation energy without altering the chemical equilibrium

Watch this video explaining how enzymes work

Author: Ruben Pio

This work is licensed under the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License. To view a copy of this license, visit http://creativecommons.org/licenses/by-nc-sa/4.0/.